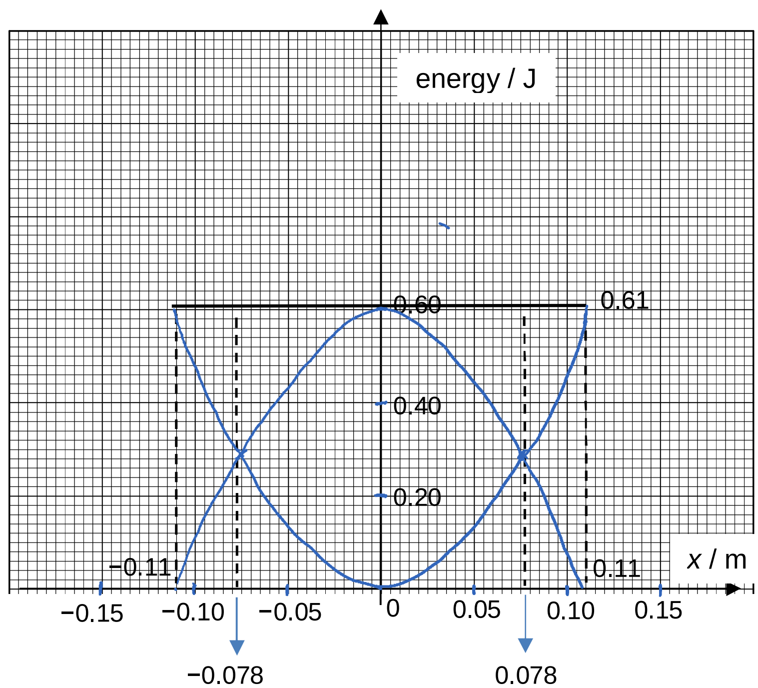
No.	Solution	Remarks
8(a)	Simple harmonic motion is an oscillatory motion in which the acceleration of an object is directly proportional to the displacement of the object from its equilibrium position, and the acceleration is always directed towards that position.	[1] for *a* proportional to *x* [1] for direction of *a*
8(b)(i)	$F_{\text{result}} = -k(e+x) + k(e-x)$ $= -ke - kx + ke - kx$ $= -2kx$ $F_{\text{result}} = ma$ $-2kx = ma$ $a = -\dfrac{2k}{m}x$ (shown)	[1] for correct equation [1] for correct expression for resultant force [1] for correct equation and substitution
8(b)(ii)	$\omega^2 = \dfrac{2k}{m}$ $\omega = \sqrt{\dfrac{2k}{m}}$ $= \sqrt{\dfrac{2(50)}{0.600}}$ $= 12.91$ $f = \dfrac{\omega}{2\pi}$ $= 2.05$ Hz	[1] for correct calculation of ω [1] for correct value of *f*
8(b)(iii)	$v_{\text{max}} = \omega x_0$ $1.4 = (12.91)x_0$ $x_0 = 0.11$ m	[1] for correct equation and substitution [1] for correct answer
8(b)(iv) 1.	Total energy, $E_T = \dfrac{1}{2}m\omega^2 x_0^2$ $= \dfrac{1}{2}(0.600)\left[\dfrac{2(50)}{0.600}\right](0.11)^2$ $= 0.61$ J	[1] for correct equation and substitution [1] for correct answer

No.	Solution	Remarks
8(b)(iv) 2.	When potential energy E_P = kinetic energy E_K, $\frac{1}{2}m\omega^2\left(x_0{}^2 - x^2\right) = \frac{1}{2}m\omega^2 x^2$ $x_0{}^2 - x^2 = x^2$ $x^2 = \frac{x_0{}^2}{2}$ $x = \frac{x_0}{\sqrt{2}}$ $= 0.078$ m	[1] for correct equation and substitution [1] for correct answer
8(b)(v)		[1] for TE line with value [1] for KE line with correct amplitude values [1] for PE line with correct intersection with KE line
8(b)(vi) 1.	If the mass of the block is larger, the angular frequency of the oscillation is lower according to the equation $\omega = \sqrt{\frac{2k}{m}}$ hence the frequency of oscillation is lower.	[1] angular frequency lower [1] frequency lower
8(b)(vi) 2.	If the surface is rough, there will be friction between the block and surface which opposes the motion of the block. This is a damping force which will cause the amplitude of the oscillation to decrease over time.	[1] friction as damping force [1] amplitude decreases over time